

Problem Set 3 *Solutions*

1. **(18 points—2 each)** In a population of students, the number of absences during the school year ranges from 0 to 8. The probabilities of a randomly drawn student from this population having 0, 1, 2, ..., 8 absences are shown in the table below. Define the event A as the student being absent *fewer than* 4 days, and the event B as the student being absent *at least* 2 days.

# of Days	0	1	2	3	4	5	6	7	8
Probability	0.2	0.14	0.25	0.11	0.1	0.09	0.05	0.03	0.03

- (a) What is the probability of event A ? $P(A) = P(0) + P(1) + P(2) + P(3) = 0.2 + 0.14 + 0.25 + 0.11 = 0.7$
- (b) What is the probability of event B ? $P(B) = P(2) + P(3) + P(4) + P(5) + P(6) + P(7) + P(8) = 0.25 + 0.11 + 0.1 + 0.09 + 0.05 + 0.03 + 0.03 = 0.66$. Or: $1 - P(0) - P(1) = 1 - 0.2 - 0.14 = 0.66$
- (c) What is the probability of $\sim A$? $P(\sim A) = 1 - P(A) = 1 - 0.7 = 0.3$
- (d) Are events A and B mutually exclusive? Explain why or why not. **No. Event A is “fewer than 4 days”; event B is “at least 2 days”. The two events intersect.**
- (e) What is the probability of $A \cap B$? $P(A \cap B) = P(2) + P(3) = 0.25 + 0.11 = 0.36$. **The outcomes of 2 and 3 absences represent the intersection of these two events.**
- (f) What is the probability of $A \cup B$? $P(A \cup B) = P(A) + P(B) - P(A \cap B) = 0.7 + 0.66 - 0.36 = 1$. **A and B represent all possible outcomes in the sample space.**
- (g) Show using values from the table that $P((A \cap B) \cup (\sim A \cap B)) = P(B)$. **In words, the lefthand side of this equation is the probability that B and A occur *or* B and $\sim A$ occur. In other words, B occurs and either A occurs or it doesn't. This is simply B, and $P(B) = 0.66$. You could also recognize that these are mutually exclusive events—if B and A are true, it cannot be the case that B and $\sim A$ are true. With mutually exclusive events, you can add the two probabilities together: $P((A \cap B) \cup (\sim A \cap B)) = 0.36 + 0.3 = 0.66$**

- (h) Show using values from the table that $P(A \cup (\sim A \cap B)) = P(A \cup B)$.
In words, the lefthand side of this equation is the probability that A occurs or A doesn't occur and B occurs. In this context (looking at the above table), this is the same as A or B occurring. As seen in part (f), this is 1.
- (i) What is $P(A|B)$? **Applying the formula $P(A|B) = P(A \cap B)/P(B)$, $P(A|B) = 0.36/0.66 = 0.545$. Notice that our “prior” probability of A, 0.70, has decreased now that we know B occurred.**

2. **(6 points—3 each)** Using the probability distribution in Question 1, find the following (and show your work):

- (a) $E(\# \text{ of absences})$:

$$\sum_{i=1}^n X_i * P(X_i) = (0 * 0.2) + (1 * 0.14) + (2 * 0.25) + (3 * 0.11) + (4 * 0.1) + (5 * 0.09) + (6 * 0.05) + (7 * 0.03) + (8 * 0.03) = \mathbf{2.57}$$

- (b) $Var(\# \text{ of absences})$ and $SD(\# \text{ of absences})$:

$$Var = \sum_{i=1}^n (X_i - E(X))^2 * P(X_i) = ((0 - 2.57)^2 * 0.20) + ((1 - 2.57)^2 * 0.14) + ((2 - 2.57)^2 * 0.25) + ((3 - 2.57)^2 * 0.11) + ((4 - 2.57)^2 * 0.1) + ((5 - 2.57)^2 * 0.09) + ((6 - 2.57)^2 * 0.05) + ((7 - 2.57)^2 * 0.03) + ((8 - 2.57)^2 * 0.03) = \mathbf{4.57}$$

$$SD = \sqrt{Var} = \sqrt{4.57} = \mathbf{2.14}$$

3. **(8 points—2 each)** Shown below is a 2 x 2 table that reports the fraction of the population in each cell:

		Education level		
		HS	<HS	Totals
Current smoker:	NO	0.614	0.130	0.744
	YES	0.194	0.062	0.256
Totals		0.808	0.192	1.000

- (a) For a randomly drawn person, what is $P(\text{smoker})$? **0.256, or 25.6%**
- (b) For a randomly drawn person, what is $P(\text{smoker} \mid <\text{HS diploma})$? **Here we can use $P(A|B) = P(A \cap B)/P(B)$, or $0.062/0.192 = 0.323$, or 32.3%**

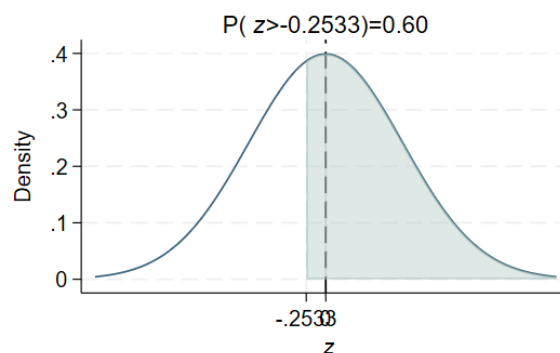
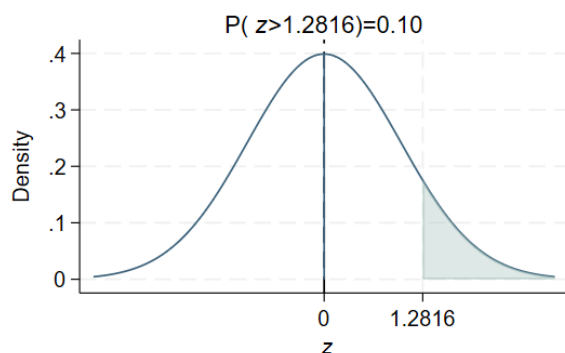
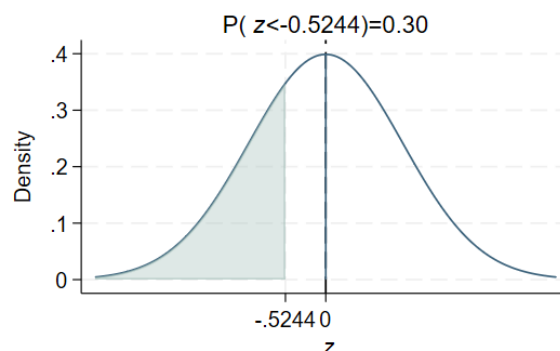
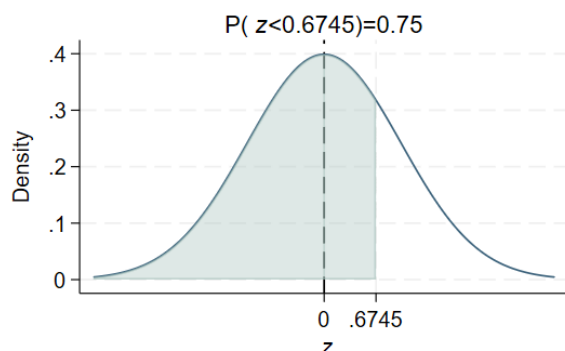
- (c) For a randomly drawn person, what is $P(\text{smoker} \mid \text{HS diploma+})$? **In the same manner as part (b): $0.194/0.808 = 0.240$, or 24.0%**
- (d) Are education and smoking status “independent?” Why or why not? **No. The probability of being a current smoker varies depending on one’s education level (as shown in parts b and c). Thus they are not independent.**
4. (5 points) After the attacks of September 11, 2001, the TSA implemented a program called SPOT (Screening of Passengers by Observation Techniques) in which passengers were flagged for suspicious behavior and given additional searching or screening. Suppose that:
- There are 2 billion plus 100 passenger trips per year (2,000,000,100).
 - 100 of these passengers are terrorists (i.e., less than 0.00000001%).
 - Nearly all (99%) terrorists exhibit the kinds of behaviors that were flagged.
 - Some non-terrorists exhibit these suspicious behaviors, but it is rare (1%).

The SPOT test has low false negative and false positive rates, suggesting it is an effective way to catch would-be terrorists. Use Bayes’ Theorem to calculate the probability that a flagged passenger is, in fact, a terrorist.

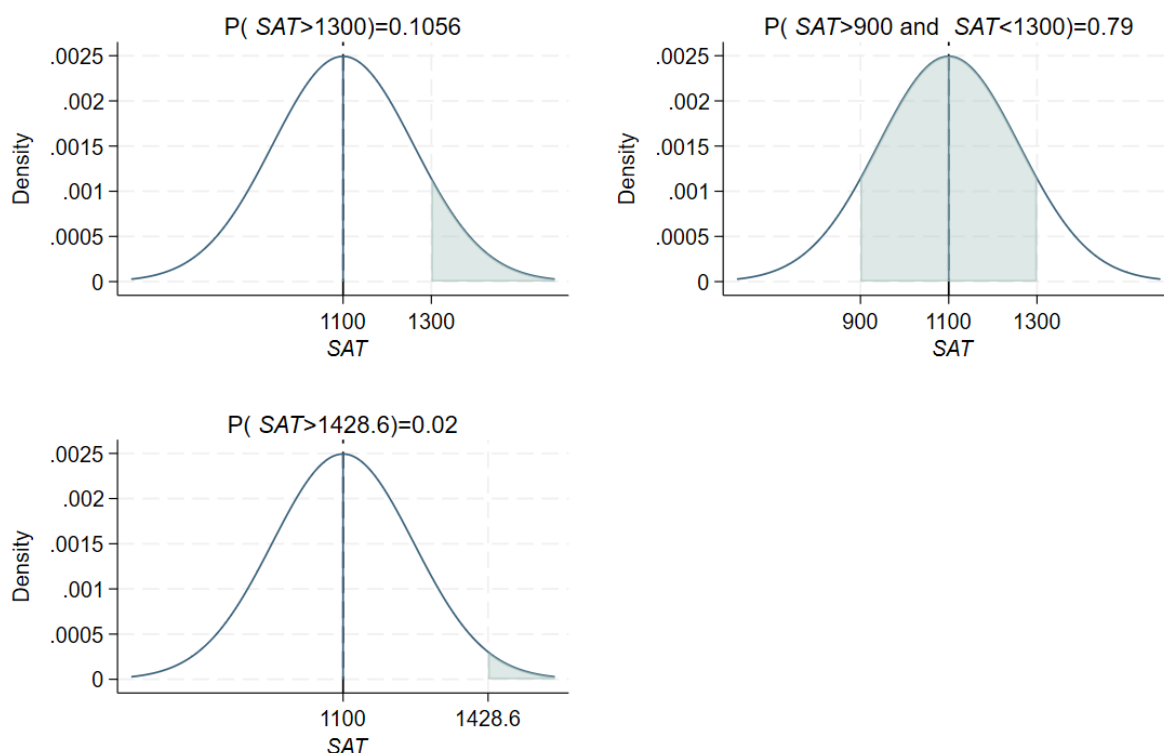
Bayes’ Theorem applied here is:

$$\begin{aligned}
 P(\text{terrorist} \mid \text{flagged}) &= \frac{P(\text{flagged} \mid \text{terrorist})P(\text{terrorist})}{P(\text{flagged})} \\
 &= \frac{\frac{99}{100} * \frac{100}{2,000,000,100}}{\frac{20,000,099}{2,000,000,100}} \\
 &= \frac{99}{20,000,099} \\
 &= 0.00000495
 \end{aligned}$$

In other words, very small! While the system involves a test that will “catch” nearly all terrorists, the baseline probability of being a terrorist is very low. Even with a low false positive rate, the SPOT system flags a very large number of innocent passengers.



5. (4 points) Assume the random variable z has a standard normal distribution. Use Stata, an online calculator, or a textbook table to answer the following:
- The probability is 0.75 that z is less than what number? $\Pr(z < 0.6745) = 0.75$,
using display `invnormal(0.75)`
 - The probability is 0.30 that z is less than what number? $\Pr(z < -0.5244) = 0.30$,
using display `invnormal(0.30)`
 - The probability is 0.10 that z is greater than what number? $\Pr(z > 1.2816) = 0.10$,
using display `(-1)*invnormal(0.10)`
 - The probability is 0.60 that z is greater than what number? $\Pr(z > -0.2533) = 0.60$,
using display `(-1)*invnormal(0.60)`
6. (6 points) Applicants to Local U. have SAT composite scores that follow a normal distribution with a mean of 1100 and a standard deviation of 160.
- What is the probability that a Local U. applicant will have a composite SAT score of 1300 or higher? $\Pr(\text{SAT} \geq 1300) = \Pr(z \geq (1300 - 1100)/160) = 0.1056$,
using display `1 - normal((1300-1100)/160)`



- (b) What is the probability that a Local U. applicant will have a composite score above 900 but below 1300? $\Pr(900 < \text{SAT} < 1300) = \Pr(z < (1300 - 1100)/160) - \Pr(z < (900 - 1100)/160) = 0.7887$, using `display normal((1300-1100)/160) - normal((900-1100)/160)`
- (c) Showing strikingly bad judgment, Local U. wishes to offer a tuition-free scholarship to applicants who score in the top 2% of applicants on the SAT. Above what score will they offer this scholarship? **The score above which 2% of applicants fall is 1428.6, found using `display 1100 + invnormal(0.98)*160`. Note the score above which 2% of applicants fall is also the score below which 98% of the applicants fall.**
7. (6 points—3 each) Suppose the probability that a teenage driver gets into an accident during a one-year period is 0.12, and assume the probability of getting into an accident is independent across drivers.
- (a) A particular family has 5 teenage drivers. What is the probability that *at least one driver* in this family will have an accident over the coming year? Show or explain how you obtained your answer.

This is an application of the binomial distribution, with 5 identical independent trials and 1 or more “successes” (getting into an accident) when the probability of “success” is 0.12.

```
. display binomialp(5,1,0.12) + binomialp(5,2,0.12) + ///
      binomialp(5,3,0.12) + binomialp(5,4,0.12) + ///
      binomialp(5,5,0.12)
.47226808

. *** or the function binomial gives you the probability of
      k or fewer successes
.
. display 1 - binomial(5,0,0.12)
.47226808
.
. *** or the function binomialtail gives you the probability of
      k or more successes
.
. display binomialtail(5,1,0.12)
.47226808
```

You could alternatively use an online calculator to find probabilities from the binomial distribution, or the formula below where $\pi = 0.12$:

$$P(x \geq k) = \sum_{i=k}^n \binom{n}{i} \pi^i (1 - \pi)^{n-i}$$

- (b) Now consider the population of families with 5 teenage drivers, and define X as the number of accidents that occurred among their 5 drivers. For these families in a typical year, what is $E(X)$ and $sd(X)$? Show or explain how you obtained your answer.

In a binomial distribution, $E(X) = n\pi$ and $Var(X) = n\pi(1 - \pi)$, so $E(X) = 5 * 0.12 = 0.6$ and $sd(X) = \sqrt{5 * 0.12 * (1 - 0.12)} = 0.727$. In other words the mean number of accidents in a year with 5 drivers is 0.6, with a standard deviation of 0.727.

8. (9 points—3 each) Simulated draws from probability distributions in Stata. Create an empty dataset with 1,000 (blank) observations
- (a) Create a variable called *score1* that consists of random draws from the normal distribution with mean 25 and standard deviation 5. After doing so, get the mean and standard deviation of *score1* and produce a histogram for it.

- (b) Create a variable called *score2* that consists of random draws from the uniform distribution between 10 and 170. After doing so, get the mean and standard deviation of *score2* and produce a histogram for it.
- (c) Challenge: create a variable called *absences* that consists of random draws from the discrete probability distribution given in part 1. Hint: how can you create a variable that is equal to 0 20% of the time, equal to 1 14% of the time, and so on? After doing so, get the mean and standard deviation of *absences* and produce a histogram for it. (This variable is discrete so you may want to use a bar graph or histogram with the discrete option).

Solution: see attached Stata log

```

.
. // *****
. // LPO.8800 Problem Set 3 - Solution to Questions 5-8
. // Last updated: September 14, 2026
. // *****
.
. // *****
. // #Q5
. // *****
.
. display invnormal(0.75)
.67448975

. display invnormal(0.30)
-.52440051

. display invnormal(1 - 0.10)
1.2815516

. display invnormal(1 - 0.60)
-.2533471

.
. // Graphs: for illustration
. // Overlay a standard normal density from -3 to 3 with
. // a shaded region above or below a z-value (p2)
.
. *** Part a
.
. global m 0
.
. global s 1
.
. global lb -3
.
. global ub 3

.
. global p1 -3
.
. global p2 0.6745

.
. # delimiter ;
delimiter now ;
. twoway function y=normalden(x,$m,$s), range($lb $ub) color(edkblue) ||
> function y=normalden(x,$m,$s), range($p1 $p2) recast(area) color(el
> tgreen%40) droplines($m)
> xtitle("{it: z}") ytitle("Density")
> title("") graphregion(fcolor(white)) name(Q5a, replace) nodraw
> subtitle("P({it: z}<0.6745)=0.75") legend(off) xlabel( $m $p2) xlin
> e($m, lpattern(dash)) ;

. # delimiter cr
delimiter now cr
.
.
. *** Part b
.

```



```

. global m 0
. global s 1
. global lb -3
. global ub 3

.
. global p1 -3
. global p2 -0.5244

.
. # delimit ;
delimiter now ;
. twoway function y=normalden(x,$m,$s), range($lb $ub) color(edkblue) droplines
> ($m) ||
> function y=normalden(x,$m,$s), range($p1 $p2) recast(area) color(el
> tgreen%40)
> xtitle("{it: z}") ytitle("Density")
> title("") graphregion(fcolor(white)) name(Q5b, replace) nodraw
> subtitle("P({it: z}<-0.5244)=0.30") legend(off) xlabel($m $p2) xli
> ne($m, lpattern(dash)) ;

. # delimit cr
delimiter now cr
.
.
. *** Part c
.
. global m 0
. global s 1
. global lb -3
. global ub 3

.
. global p1 1.2816
. global p2 3

.
. # delimit ;
delimiter now ;
. twoway function y=normalden(x,$m,$s), range($lb $ub) color(edkblue) droplines
> ($m) ||
> function y=normalden(x,$m,$s), range($p1 $p2) recast(area) color(el
> tgreen%40)
> xtitle("{it: z}") ytitle("Density")
> title("") graphregion(fcolor(white)) name(Q5c, replace) nodraw
> subtitle("P({it: z}>1.2816)=0.10") legend(off) xlabel($p1 $m ) xlin
> e($m, lpattern(dash)) ;

. # delimit cr
delimiter now cr
.
.

```

```

. *** Part d
.
. global m 0
. global s 1
. global lb -3
. global ub 3
.
. global p1 -0.2533
. global p2 3
.
. # delimit ;
delimiter now ;
. twoway function y=normalden(x,$m,$s), range($lb $ub) color(edkblue) ||
>          function y=normalden(x,$m,$s), range($p1 $p2) recast(area) color(el
> tgreen%40) droplines($m)
>          xtitle("{it: z}") ytitle("Density")
>          title("") graphregion(fcolor(white)) name(Q5d, replace) nodraw
>          subtitle("P({it: z}>-0.2533)=0.60") legend(off) xlabel($p1 $m ) xli
> ne($m, lpattern(dash)) ;

. # delimit cr
delimiter now cr
.
. graph combine Q5a Q5b Q5c Q5d, rows(2) graphregion(fcolor(white) lcolor(none))

. graph export PS3-Q5.png, as(png) replace
file PS3-Q5.png saved as PNG format

. graph export PS3-Q5.pdf, as(pdf) replace
file PS3-Q5.pdf saved as PDF format

.
.
. // *****
. // #6
. // *****
.
. *** part a
.
. display 1 - normal((1300-1100)/160)
.10564977

.
. *** part b
.
. display normal((1300-1100)/160) - normal((900-1100)/160)
.78870045

.
. *** part c
.
. scalar z = invnormal(0.98)

. display z
2.0537489

```

```

. display 1100 + z*160
1428.5998

.
.
. // Graphs for illuatration
.
. *** Part a
.
. global m 1100
.
. global s 160
.
. global lb = $m - (3*$s)
.
. global ub = $m + (3*$s)
.
.
. global p1 = 1300
. global p2 = $ub
.
. # delimit ;
delimiter now ;
. twoway function y=normalden(x,$m,$s), range($lb $ub) color(edkblue) droplines(
> $m) ||
> function y=normalden(x,$m,$s), range($p1 $p2) recast(area) color(el
> tgreen%40)
> xtitle("{it: SAT}") ytitle("Density")
> title("") graphregion(fcolor(white)) name(Q6a, replace)
> subtitle("P({it: SAT}>1300)=0.1056") legend(off) xlabel( $p1 $m) x1
> ine($m, lpattern(dash)) ;

. # delimit cr
delimiter now cr
.
.
. *** Part b
.
. global m 1100
.
. global s 160
.
. global lb = $m - (3*$s)
.
. global ub = $m + (3*$s)
.
.
. global p1 = 900
. global p2 = 1300
.
. # delimit ;
delimiter now ;
. twoway function y=normalden(x,$m,$s), range($lb $ub) color(edkblue) droplines
> ($m) ||
> function y=normalden(x,$m,$s), range($p1 $p2) recast(area) color(el
> tgreen%40)
> xtitle("{it: SAT}") ytitle("Density")
> title("") graphregion(fcolor(white)) name(Q6b, replace)
> subtitle("P({it: SAT}>900 and {it: SAT}<1300)=0.79") legend(off) x1
> abel( $p1 $p2 $m ) xline($m, lpattern(dash)) ;

```

```

. # delimit cr
delimiter now cr
.
. *** Part c
.
. global m 1100
. global s 160
. global lb = $m - (3*$s)
. global ub = $m + (3*$s)
.
. global p1 1428.6
. global p2 = $ub
.
. # delimit ;
delimiter now ;
. twoway function y=normalden(x,$m,$s), range($lb $ub) color(edkblue) droplines
> ($m) ||
> function y=normalden(x,$m,$s), range($p1 $p2) recast(area) color(el
> tgreen%40)
> xtitle("{it: SAT}") ytitle("Density")
> title("") graphregion(fcolor(white)) name(Q6c, replace)
> subtitle("P({it: SAT}>1428.6)=0.02") legend(off) xlabel($p1 $m ) x1
> ine($m, lpattern(dash)) ;
.
. # delimit cr
delimiter now cr
.
.
. graph combine Q6a Q6b Q6c , rows(2) graphregion(fcolor(white) lcolor(none))
. graph export PS3-Q6.png, as(png) replace
file PS3-Q6.png saved as PNG format
. graph export PS3-Q6.pdf, as(pdf) replace
file PS3-Q6.pdf saved as PDF format
.
.
. // *****
. // #7
. // *****
.
. *** Part a
. *** This is a binomial distribution problem, with 5 identical independent tria
> ls
. *** and 1 or more "successes" (getting into an accident) when the probability
> of
. *** "success" is 0.12
.
. display binomialp(5,1,0.12) + binomialp(5,2,0.12) + binomialp(5,3,0.12) + ///
> binomialp(5,4,0.12) + binomialp(5,5,0.12)
.47226808

```

```

. *** or the function binomial gives you the probability of k or fewer successes
.
. display 1 - binomial(5,0,0.12)
.47226808

.
. *** or the function binomialtail gives you the probability of k or more successes
> ses
.
. display binomialtail(5,1,0.12)
.47226808

.
.
. // *****
. // #8
. // *****
.
. clear

. set obs 1000
Number of observations (_N) was 0, now 1,000.

.
. gen score1=rnormal(25,5)

. summ score1

  Variable |      Obs      Mean   Std. dev.      Min      Max
-----+-----
  score1 |    1,000    24.94018     4.9853   9.984358   39.45221

. histogram score1, name(Q8a, replace)
(bin=29, start=9.9843578, width=1.0161328)

.
. gen score2=runiform(10,170)

. summ score2

  Variable |      Obs      Mean   Std. dev.      Min      Max
-----+-----
  score2 |    1,000    90.45146    45.99532   10.06611   169.8053

. histogram score2, name(Q8b, replace)
(bin=29, start=10.066107, width=5.5082474)

.
. gen randu = runiform(0,1)

. gen absences = 0 if randu<0.2
(799 missing values generated)

. replace absences = 1 if absences==. & randu<0.34
(134 real changes made)

. replace absences = 2 if absences==. & randu<0.59
(263 real changes made)

```

```

. replace absences = 3 if absences==. & randu<0.70
(95 real changes made)

. replace absences = 4 if absences==. & randu<0.80
(108 real changes made)

. replace absences = 5 if absences==. & randu<0.89
(86 real changes made)

. replace absences = 6 if absences==. & randu<0.94
(56 real changes made)

. replace absences = 7 if absences==. & randu<0.97
(32 real changes made)

. replace absences = 8 if absences==. & randu<=1
(25 real changes made)

. fre absences

```

absences

		Freq.	Percent	Valid	Cum.
Valid	0	201	20.10	20.10	20.10
	1	134	13.40	13.40	33.50
	2	263	26.30	26.30	59.80
	3	95	9.50	9.50	69.30
	4	108	10.80	10.80	80.10
	5	86	8.60	8.60	88.70
	6	56	5.60	5.60	94.30
	7	32	3.20	3.20	97.50
	8	25	2.50	2.50	100.00
	Total	1000	100.00	100.00	

```

.
. summ absences

```

Variable	Obs	Mean	Std. dev.	Min	Max
absences	1,000	2.567	2.125566	0	8

```

. histogram absences, discrete name(Q8c, replace)
(start=0, width=1)

```

```

.
. graph combine Q8a Q8b Q8c, ysize(6) xsize(7) rows(2)

```

```

. graph export PS3-Q8.png, as(png) replace
file PS3-Q8.png saved as PNG format

```

```

. graph export PS3-Q8.pdf, as(pdf) replace
file PS3-Q8.pdf saved as PDF format

```

```

.
. // Close log and convert to PDF
. log close

```

